

PHYSICS-INFORMED BUT NOT PHYSICS-CONSISTENT: ERROR GEOMETRY AND SUBSPACE PROJECTION FOR NEURAL AC POWER FLOW

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ABSTRACT

Recent neural power-flow solvers, including emerging foundation models, achieve accurate voltage predictions, yet such accuracy does not necessarily imply physically consistent solutions. Even small voltage errors can yield large residuals from the AC power-balance. We study this accuracy–consistency gap across PIGNN, GridSFM, gridfm-graphkit, and LUMINA on realistic 2224-bus GBnetwork scenarios, with cross-grid evaluation of GridSFM over 31 systems. Using an SVD basis fitted to training AC power-flow solutions, we find that neural prediction errors contain substantial components outside the dominant solution subspace. To address this mismatch, calibrated solution-subspace projection (CSP) removes off-subspace prediction components, substantially reducing mean power-balance residual from 0.3159 to 0.0550 p.u. for PIGNN-GC and from 0.2687 to 0.0514 p.u. for GridSFM. These results highlight output-error geometry as an important factor in physics-consistent neural AC power flow.

Index Terms— AC power flow, neural surrogate, physics violation, error geometry, calibration, distribution shift

1. INTRODUCTION

Modern power-system operation requires repeated AC power-flow (AC-PF) evaluations across increasingly diverse operating conditions driven by renewable generation, electrification, and changing demand [1, 2]. AC-PF maps network conditions and power injections to bus voltages and flows serving as a fundamental nonlinear primitive in power-system analysis [3]. Repeated evaluations arise in time-series and contingency studies [4], while the same AC network equations underlie optimization and state-estimation tasks [5].

The computational demands of repeated grid analysis have motivated learned surrogates that amortize the cost of conventional solvers over many operating points [6]. Learned grid solvers span different levels of constraint enforcement: direct regression prioritizes fast inference [7], physics- or constraint-aware objectives penalize violations during training [8, 9], while iterative correction and feasibility-seeking mechanisms refine predictions more directly toward physically consistent [10, 11] or feasible states [12, 13]. More recent cross-grid and foundation-model efforts—including GridSFM [14], gridfm-graphkit and GENCO [15, 16], and LUMINA [17]—extend

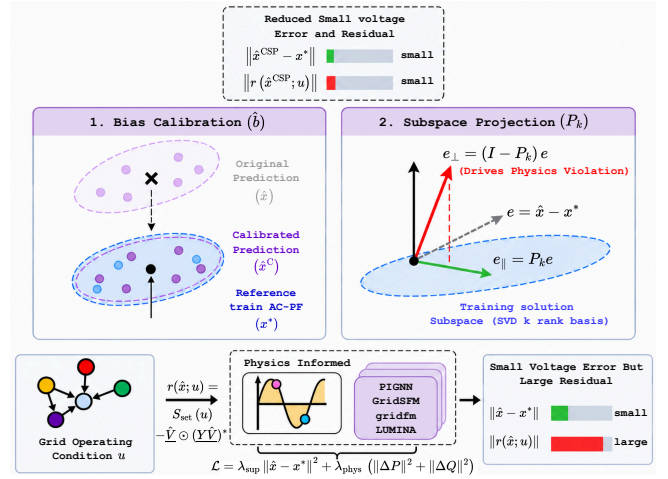


Fig. 1. Neural AC-PF surrogates can achieve small voltage errors while retaining large physics violations. Bias calibration and solution-subspace projection jointly suppress systematic and off-subspace errors, improving both voltage accuracy and AC power-balance consistency.

this direction with global context and rich heterogeneous, topology-aware representations designed to operate across diverse grids and operating conditions.

Despite these advances, accurate voltage prediction does not necessarily imply AC-consistent output [18]. Physics-informed objectives can substantially reduce power-flow residuals [19, 18], but residual penalties remain soft training objectives and do not by themselves enforce exact satisfaction of the nonlinear AC equations. Consequently, small voltage-state errors can coexist with non-negligible power-balance violations. This raises a question not addressed by state accuracy alone: what properties of the remaining voltage error determine how strongly it is translated into AC power-balance residuals?

We hypothesize that the missing factor is the *direction* of the remaining voltage error: errors of similar magnitude may produce very different power-balance residuals depending on how they are oriented across buses. Goodwin et al. [20] characterized the differential geometry of the AC-PF solution manifold and assessed the quality of its linear approximations. For neural surrogates, we instead study error geometry empirically, asking whether the orientation of voltage errors relative to a linear subspace estimated from training solutions

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helps explain the remaining physical inconsistency.

We evaluate this hypothesis across PIGNN-GC, GridSFM, gridfm-graphkit, and LUMINA on realistic 2224-bus GBnetwork scenarios, with additional cross-grid evaluation of GridSFM over 31 systems of diverse sizes. Across all four models, train-only calibration improves voltage accuracy while leaving persistent power-balance residuals, showing that persistent bias alone does not explain the accuracy–consistency gap. We therefore construct an SVD basis exclusively from training AC power-flow solutions and examine the orientation of the remaining voltage errors relative to the resulting training-supported solution subspace. Unlike Park et al., who use a PCA-based subspace to compress and reconstruct AC-OPF outputs during learning [21], we use the training-solution subspace to analyze surrogate-error geometry and to correct frozen AC-PF predictions post hoc.

We find that substantial prediction error can lie outside the dominant training-supported solution directions and that suppressing this off-subspace component markedly reduces AC power-balance violations. Motivated by this observation, we introduce *calibrated solution-subspace projection* (CSP), which first removes train-estimated bias and then suppresses prediction components outside the dominant training-supported directions. CSP improves voltage-magnitude accuracy and AC power balance simultaneously, while cross-grid and distribution-shift experiments show that its benefits extend across diverse systems but depend on how well the training-derived subspace represents the operating distribution. Together, these results identify voltage-error geometry across buses as an important link between state accuracy and physical consistency.

Code: github.com/Kimchangheon/neural-acpf-error-geometry

2. ERROR GEOMETRY AND CALIBRATED SUBSPACE PROJECTION

2.1. AC power-balance consistency and train-only calibration

We consider an N -bus grid with voltage state $x = [V, \theta] \in \mathbb{R}^{2N}$, where $V = V \odot e^{j\theta}$. Given an operating condition u , a neural surrogate predicts $\hat{x} = f_\phi(u)$, from which the AC active and reactive powers (P, Q) are recomputed using the network admittance model. For setpoints $(P^{\text{set}}, Q^{\text{set}})$, define

$$r(\hat{x}; u) = \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} P^{\text{set}} - P \\ Q^{\text{set}} - Q \end{bmatrix}, \quad \text{PB} = \sqrt{\Delta P^2 + \Delta Q^2}.$$

AC power-balance consistency is assessed by PB (Power Balance), evaluated bus-wise under the standard slack/PV/PQ conventions; Mean PB and Max PB are reported over buses and scenarios. Before geometric analysis, we remove train-only post-training bias:

$$\hat{b} = \mathbb{E}_{s \in \text{tr}} [\hat{x}_s - x_s^*], \quad \hat{x}^C = \hat{x} - \hat{b},$$

with $\hat{b}$ frozen at test time. Angle offsets use circular differences after gauge alignment; calibration removes bias without constraining residual error directions.

2.2. Error geometry in a training-derived solution subspace

To characterize the remaining error, we construct a rank- k -per-channel basis from Newton–Raphson solutions in the training set. Let X_V and X_θ denote the centered training matrices of voltage magnitudes and gauge-aligned angles, respectively. We compute separate SVDs

$$X_V = W_V \Sigma_V U_V^\top, \quad X_\theta = W_\theta \Sigma_\theta U_\theta^\top, \quad P_k = W_k W_k^\top, \quad (1)$$

where $W_k = \text{diag}(W_{V,k}, W_{\theta,k})$ contains the leading k directions from each SVD; k is capped by the available components on smaller systems.

These subspaces capture dominant training-solution variation and define our error coordinates. For $e = \hat{x} - x^*$, we decompose

$$e^\parallel = P_k e, \quad e^\perp = (I - P_k) e, \quad (2)$$

where $e^\parallel$ lies along the dominant training-solution directions and $e^\perp$ captures error outside them.

To relate this error decomposition to physical consistency, we use a first-order Taylor expansion of the AC residual around a converged power-flow solution x^* :

$$r(x^* + e; u) = \underbrace{r(x^*; u)}_{=0} + \underbrace{J(x^*; u)e}_{\text{first-order term}} + \underbrace{\mathcal{O}(\|e\|^2)}_{\text{higher-order terms}}, \quad (3)$$

where J is the AC residual Jacobian. Thus, the Jacobian determines how a given voltage error is translated into power-balance residual: errors aligned with directions of larger Jacobian gain can produce substantially larger residuals even when their magnitudes are similar. This motivates testing whether AC power balance changes when the voltage error is rotated toward or away from the training-derived solution subspace while its magnitude is held fixed.

2.3. Calibrated solution-subspace projection (CSP)

This motivates a direct intervention: suppress prediction components outside the dominant training-solution directions while preserving the variation supported by the data. Because systematic bias and off-subspace error are distinct, we first apply post-training calibration and then project the calibrated prediction:

$$\hat{x}^{\text{CSP}} = \bar{x}_{\text{tr}} + P_k \left(\hat{x} - \hat{b} - \bar{x}_{\text{tr}} \right) \quad (4)$$

where $\hat{b}$ is estimated from the trained model’s predictions on the training set and frozen at test time, while $\bar{x}_{\text{tr}}$ and P_k are obtained solely from the training reference solutions.

The projection removes off-subspace prediction error, but can also discard valid reference variation. To characterize this trade-off, let $e^C = \hat{x}^C - x^*$. Since P_k is orthogonal,

$$\|e^{\text{CSP}}\|_2^2 - \|e^C\|_2^2 = \|(I - P_k)(x^* - \bar{x}_{\text{tr}})\|_2^2 - \|(I - P_k)e^C\|_2^2. \quad (5)$$

Thus, CSP improves state accuracy when the removed off-subspace prediction error exceeds the valid reference variation discarded by the projection.

CSP is a data-driven output correction rather than a projection onto the nonlinear AC-feasible set, and therefore does not guarantee exact satisfaction of the AC equations or operational constraints. Instead, it tests whether suppressing off-subspace prediction components improves voltage accuracy and recomputed AC power-balance residuals.

3. EXPERIMENTS AND RESULTS

3.1. Experimental setup

Our primary study uses 37,022 converged pandapower Newton–Raphson operating points on the 2224-bus GBnetwork, split 1:1:1 for training/validation/test and varying load, generation, and PV setpoints [22]. We evaluate PIGNN-GC, our global-context extension

of PIGNN-Attn-LS [23], GridSFM [14], gridfm-graphkit v0.9.0 [15], and LUMINA [17]; GridSFM is additionally evaluated cross-grid on 31 systems.

All models are trained from scratch using model-specific stable initialization, and CSP is evaluated within each frozen model. gridfm-graphkit uses its released architecture/configuration without a pretrained checkpoint. GridSFM retains the released backbone but is adapted from AC-OPF to AC-PF by fixing generator dispatch and training only for voltage-state prediction with PF-specific residuals. LUMINA likewise retains its released backbone but uses an enhanced AC-PF configuration with known-variable masking, electrical bus features, and stabilized initialization. For all models, electrical inputs, admittance, and evaluation use the same pandapower internal PPC representation [24, 25].

All calibration statistics and solution bases are fitted on training data only. From $k \in \{4, 8, 16, 32, 64\}$, validation Mean PB selects $k^* = 16$ without degrading calibrated voltage RMSE; this rank is then frozen for all test and distribution-shift experiments.

Voltage RMSE alone can be small even when a model largely predicts a static per-bus profile rather than following scenario-dependent voltage variation [14]. To isolate this scenario-dependent response, we center each bus as $\tilde{V}_{is} = V_{is} - \bar{V}_i$ and $\tilde{V}_{is} = \hat{V}_{is} - \bar{\hat{V}}_i$, and report

$$R_w^2 = \text{corr}^2(\tilde{V}_{is}, \tilde{\hat{V}}_{is}), \quad a_w = \frac{\text{Cov}(\tilde{V}_{is}, \tilde{\hat{V}}_{is})}{\text{Var}(\tilde{\hat{V}}_{is})}. \quad (6)$$

R_w^2 is the squared correlation between the centered prediction and reference, while a_w measures the magnitude of the predicted variation relative to the true variation.

3.2. Fixed-norm directional intervention.

We vary the orientation of the voltage-magnitude error relative to the training-derived solution subspace while holding its ℓ_2 norm fixed, thereby isolating direction from error magnitude. Starting from the calibrated prediction, we define

$$\tilde{e}(\alpha) = \|e_V\|_2 \frac{e_V^\parallel + \alpha e_V^\perp}{\|e_V^\parallel + \alpha e_V^\perp\|_2}. \quad (7)$$

Decreasing α suppresses off-subspace error; the control analogously suppresses the in-subspace component by exchanging $e_V^\parallel$ and $e_V^\perp$. Phase angles are held fixed and Mean PB is normalized at $\alpha = 1$. Across all four surrogates, suppressing off-subspace error consistently reduces Relative Mean PB, whereas suppressing in-subspace error produces the opposite trend (Fig. 2). Although the effect size varies across models, its direction is consistent. Because the voltage-magnitude error norm is fixed throughout the intervention, these results show that error orientation affects AC power-balance violation independently of error magnitude.

Consistent with the fixed-norm experiment, a complementary local Jacobian analysis shows that off-subspace perturbations produce larger AC-residual changes per unit perturbation norm than in-subspace perturbations. In a model-independent audit of 4,096 paired random directions at 128 held-out operating points, the median ratio of off- to in-subspace Jacobian gain is $1,563\times$. The same directional sensitivity is present in the actual calibrated prediction errors: the corresponding off-/in-subspace gain ratios are $22.81\times$, $575.83\times$, $4.69\times$, and $6.51\times$ for PIGNN-GC, GridSFM, gridfm-graphkit, and LUMINA, respectively. The first-order expansion in Eq. (3) remains

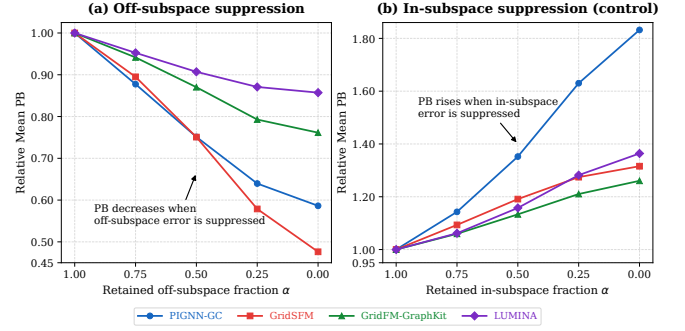


Fig. 2. Fixed-norm directional intervention on GBnetwork, with Relative Mean PB normalized at $\alpha = 1$.

Table 1. Basis controls on GBnetwork for PIGNN-GC (seed 42, $k = 16$). Random denotes the mean over eight independently sampled bases.

Projection	$ V $ RMSE $\downarrow$	Angle RMSE ($^\circ$) $\downarrow$	R_w^2 $\uparrow$	a_w	Mean PB $\downarrow$
None (C)	.01421	.753	.68522	.49329	.14511
Solution SVD	.01287	.822	.85299	.48903	.05690
Random	.02336	6.825	.00540	.00290	17.74
Laplacian LF	.02132	5.901	.20128	.12304	30.53
Spatial LP	.01568	1.926	.63080	.40717	29.29

accurate in the observed error regime, with median linearized-to-nonlinear residual-change ratios between 0.996 and 1.009 across all four models.

3.3. Specificity of the solution subspace.

To test whether the observed benefit follows from generic dimensionality reduction or graph smoothing, we replace the solution-SVD projection in CSP with matched random, graph-Laplacian, and spatial low-pass projections. As shown in Table 1, only the training-solution SVD improves both voltage-magnitude accuracy and power balance while increasing R_w^2 from .685 to .853 and nearly preserving a_w . In contrast, all three controls degrade scenario tracking and substantially increase AC residuals. Thus, the CSP gain cannot be explained by low dimensionality or spatial smoothing alone, but depends on the orientation of the training-solution subspace.

3.4. Post-training CSP correction

Table 2 compares raw predictions, calibration (C), projection (P_{16}), and CSP₁₆ on frozen outputs from PIGNN-GC, GridSFM, gridfm-graphkit, and LUMINA. Relative to C , CSP reduces Mean PB by 67.0%, 37.8%, 40.5%, and 71.1%, respectively, while lowering voltage-magnitude RMSE in all four models. Projection also increases R_w^2 with little change in a_w . Angle accuracy is model dependent, degrading for the first three models but improving for LUMINA. Raw and CSP₁₆ GPU inference times remain comparable across all four models, indicating no discernible systematic overhead from the post-training correction.

Comparing magnitude-only, angle-only, and full projection reveals that the contribution of angle projection is model dependent. For PIGNN-GC, full CSP gives the lowest PB among the projection variants, whereas GridSFM achieves its lowest PB with magnitude-only projection. For gridfm-graphkit, adding angle projec-

tion to the magnitude-only correction reduces Mean/Max PB from .0612/18.23 to .0457/5.08, but increases angle RMSE from .367° to .496°. For LUMINA, the same addition reduces Mean/Max PB from .1939/38.53 to .0660/4.51 while also improving angle RMSE from 1.374° to 1.215°. Thus, improved physical consistency from angle projection does not necessarily coincide with improved angle accuracy.

These model-dependent accuracy changes are explained by Eq. (5). For magnitude, the removed off-subspace prediction-error energy exceeds the discarded reference variation across all four models. For angle, the balance differs across models: in gridfm-graphkit, angle projection discards more valid reference variation than model error and therefore worsens angle RMSE, whereas the opposite holds for LUMINA.

3.5. Complementarity with one-step Newton refinement.

We next test whether CSP remains useful when followed by an explicit residual-directed correction. Starting from either calibrated outputs (C) or CSP_{16} , we apply exactly one Newton update (NR_1), updating non-slack angles and PQ voltage magnitudes. The damping factor is selected on validation data from $\{0.25, 0.5, 0.75, 1.0\}$ and fixed to $\eta = 1$ for all models. No Newton solve failures or damping fallbacks occur.

A single Newton step substantially improves both voltage accuracy and power balance from either initialization. Importantly, the benefit of CSP persists after this strong correction: relative to $C + NR_1$, $CSP_{16} + NR_1$ reduces Mean PB by 47.9%, 51.2%, 21.1%, and 81.0% for PIGNN-GC, GridSFM, gridfm-graphkit, and LUMINA, respectively, while also lowering magnitude and angle errors in all four models. Thus, CSP and Newton refinement address complementary aspects of the prediction error: CSP provides a data-derived geometric correction, whereas NR_1 explicitly uses the AC Jacobian to reduce the remaining residual.

The computational costs are correspondingly different. CSP requires approximately two seconds for the full 12,344-scenario test set, whereas one scenario-wise Newton correction requires approximately $2.4\text{--}3.8 \times 10^3$ s in our multiprocessing implementation. These implementation-specific timings exclude neural inference and are not hardware-normalized. CSP is therefore not a replacement for Newton refinement, but a lightweight correction that also provides a better initialization for a subsequent residual-directed update.

3.6. Cross-grid breadth and distribution shift

Cross-grid breadth. Fig. 3 evaluates GridSFM with CSP across 31 systems spanning 4–9,241 buses and sub-kV to 750-kV levels: 28 pandapower/PYPOWER-format systems from IEEE, PEGASE, RTE, and GB benchmark families, plus three Common Information Model (CIM)-based networks. For each system, calibration and subspace statistics are fitted from its training data, using nominal $k = 16$ per channel and the available SVD components on smaller systems. CSP reduces GridSFM Mean PB across all 31 systems.

Topology and spatial shifts. Under $N-1/N-2$ outages, tracking and bulk PB remain close to control, with larger maxima indicating isolated topology-induced violations and the need for worst-case assessment beyond average error [26]. GBcorr causes a broader shift: transferring both control calibration and basis gives Mean PB ≈ 0.100 p.u. Refitting calibration alone reduces this to 0.0944, basis alone to 0.0740, and both to ≈ 0.068 . The transferred basis retains 93.55% of GBcorr magnitude variation, versus 98.53% after refitting, while reference-state projection reduces Mean PB from 0.0868 to 0.0438

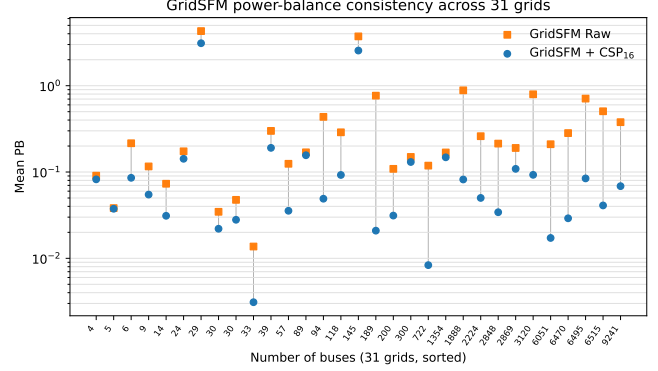


Fig. 3. Grid-by-grid physical consistency of GridSFM before and after CSP_{16} across 31 systems ordered by bus count. Mean PB is shown on a logarithmic scale.

p.u. This shows that GBcorr introduces not only calibration drift but also solution-subspace mismatch.

4. CONCLUSION

This work identifies output-error geometry as an important factor in the accuracy–consistency gap of neural AC-PF surrogates. Fixed-norm and Jacobian analyses reveal strong directional sensitivity of AC residuals, while CSP reduces power-balance residuals and improves voltage-magnitude accuracy on GBnetwork. CSP also improves voltage accuracy and mean power-balance residuals after one Newton update, supporting its complementary role in residual-directed refinement. Cross-grid evaluation shows that the power-balance improvement extends beyond GBnetwork, whereas distribution-shift experiments show that a fixed solution subspace can become mismatched to the shifted operating distribution. These results motivate extending physics-consistent neural surrogates toward broader power-system analysis, estimation, and optimization.

5. REFERENCES

- [1] Giovanni De Carne, S. Masoome Maroufi, Hamzeh Beiranvand, Valerio De Angelis, Salvatore D’Arco, Vahan Gevorgian, Simon Waczowicz, Barry Mather, Marco Liserre, and Veit Hagenmeyer, “The role of energy storage systems for a secure energy supply: A comprehensive review of system needs and technology solutions,” *Electric Power Systems Research*, vol. 236, pp. 110963, 2024.
- [2] Timon Conrad, Changhun Kim, Johann Jäger, Andreas Maier, and Siming Bayer, “Impact of training dataset size for ML load flow surrogates,” in *Proceedings of the Oberlausitzer Energiesymposium 2025 & Zittauer Energieseminar*, Zittau, Germany, 2025, arXiv:2602.19667.
- [3] M. Karimi, A. Shahriari, M. R. Aghamohammadi, H. Marzoghi, and V. Terzija, “Application of newton-based load flow methods for determining steady-state condition of well and ill-conditioned power systems: A review,” *International Journal of Electrical Power & Energy Systems*, vol. 113, pp. 298–309, 2019.
- [4] Muhammed Öz, Jasmin Hörter, Kaleb Phipps, Charlotte Debus,

Table 2. GBnetwork results at $k = 16$. PIGNN-GC, GridSFM, and gridfm-graphkit report mean $\pm$ standard deviation over three model seeds; LUMINA is reported for seed 42. C denotes calibration, P_{16} projection, and CSP_{16} their composition. Inference time is GPU-synchronized forward execution plus the corresponding output transform, reported per scenario.

Model	Output	$ V $ RMSE $\downarrow$	Angle RMSE ($^\circ$) $\downarrow$	R_w^2 $\uparrow$	a_w	Mean PB $\downarrow$	Max PB $\downarrow$	Inference (ms/scen.)
Per-bus mean	Baseline	.02341	6.847	.000	.000	.12084	5.88	–
PIGNN-GC	Raw	.02850 $\pm$.00232	1.108 $\pm$.175	.671 $\pm$.050	.546 $\pm$.072	.31592 $\pm$.03240	195.89 $\pm$ 52.54	22.031 $\pm$.209
	C	.01393 $\pm$.00105	.777 $\pm$.161	.671 $\pm$.050	.546 $\pm$.072	.16661 $\pm$.02352	190.00 $\pm$ 37.53	22.045 $\pm$.255
	P_{16}	.01624 $\pm$.00254	1.091 $\pm$.166	.829 $\pm$.045	.541 $\pm$.071	.07594 $\pm$.00422	5.07 $\pm$.85	21.954 $\pm$.127
	CSP_{16}	.01222 $\pm$.00134	.835 $\pm$.141	.829 $\pm$.045	.541 $\pm$.071	.05497 $\pm$.00257	3.93 $\pm$.07	22.046 $\pm$.206
GridSFM	Raw	.02868 $\pm$.01652	2.294 $\pm$.302	.768 $\pm$.151	.724 $\pm$.149	.26867 $\pm$.13004	9.10 $\pm$ 4.02	25.045 $\pm$.207
	C	.01086 $\pm$.00403	1.781 $\pm$.055	.768 $\pm$.151	.724 $\pm$.149	.08251 $\pm$.01093	7.29 $\pm$ 3.53	25.015 $\pm$.096
	P_{16}	.01814 $\pm$.01145	2.244 $\pm$.325	.936 $\pm$.032	.724 $\pm$.149	.07630 $\pm$.01942	5.08 $\pm$.49	25.087 $\pm$.279
	CSP_{16}	.00782 $\pm$.00307	1.807 $\pm$.053	.936 $\pm$.032	.724 $\pm$.149	.05136 $\pm$.00143	4.94 $\pm$.42	24.577 $\pm$.091
gridfm-graphkit	Raw	.00892 $\pm$.00154	.385 $\pm$.042	.905 $\pm$.030	.811 $\pm$.045	.09282 $\pm$.01763	19.82 $\pm$ 8.65	5.917 $\pm$.019
	C	.00753 $\pm$.00118	.367 $\pm$.042	.905 $\pm$.030	.811 $\pm$.045	.07685 $\pm$.01307	18.88 $\pm$ 8.69	5.730 $\pm$.004
	P_{16}	.00617 $\pm$.00099	.503 $\pm$.018	.959 $\pm$.008	.801 $\pm$.044	.04593 $\pm$.00156	5.36 $\pm$.56	5.723 $\pm$.011
	CSP_{16}	.00606 $\pm$.00090	.496 $\pm$.021	.959 $\pm$.008	.801 $\pm$.044	.04571 $\pm$.00135	5.08 $\pm$.18	5.779 $\pm$.012
LUMINA	Raw	.02689	2.347	.448	.358	.83072	464.63	8.469
	C	.01768	1.373	.448	.358	.22800	38.85	8.791
	P_{16}	.01943	1.856	.892	.350	.07713	4.83	8.121
	CSP_{16}	.01549	1.215	.892	.350	.06597	4.51	8.739

Table 3. One Newton update from calibrated or CSP_{16} outputs on GBnetwork. Validation selects $\eta = 1$ for all models.

Model	Start	$ V $ RMSE $\downarrow$	Angle RMSE ($^\circ$) $\downarrow$	R_w^2 $\uparrow$	a_w	Mean PB $\downarrow$	Max PB $\downarrow$
PIGNN-GC	C	.00180 $\pm$.00012	.119 $\pm$.013	.99705 $\pm$.00030	.975 $\pm$.001	.00284 $\pm$.00046	1.72 $\pm$.82
	CSP_{16}	.00117 $\pm$.00010	.073 $\pm$.009	.99875 $\pm$.00021	.979 $\pm$.002	.00148 $\pm$.00013	.782 $\pm$.021
GridSFM	C	.01186 $\pm$.00677	.681 $\pm$.380	.95000 $\pm$.02347	1.385 $\pm$.266	.00250 $\pm$.00154	.860 $\pm$.255
	CSP_{16}	.00097 $\pm$.00020	.059 $\pm$.013	.99912 $\pm$.00038	.983 $\pm$.003	.00122 $\pm$.00029	.848 $\pm$.093
gridfm-graphkit	C	.000943 $\pm$.000232	.0843 $\pm$.0236	.99922 $\pm$.00037	.98910 $\pm$.00208	.000941 $\pm$.000325	1.112 $\pm$.549
	CSP_{16}	.000496 $\pm$.000085	.0277 $\pm$.0041	.99980 $\pm$.00008	.99033 $\pm$.00126	.000742 $\pm$.000080	.624 $\pm$.080
LUMINA	C	.008687	.6287	.93290	.93327	.008936	2.474
	CSP_{16}	.001057	.0838	.99910	.98270	.001694	.721

Table 4. Fixed-rank CSP_{16} under topology and operating-distribution shifts. T transfers the control-derived calibration and solution basis, whereas R refits both on condition-matched training data. PB statistics are pooled over all bus–scenario pairs.

Condition	RMSE $_V$	R_w^2 $\uparrow$	a_w	Mean	Median	p95	p99	Max
Control	.01281	.8476	.4866	.0581	.0131	.2665	.5462	3.58
$N-1$ (T)	.01314	.8391	.4750	.0603	.0131	.2713	.5632	33.16
$N-2$ (T)	.01288	.8458	.4870	.0617	.0132	.2713	.5721	27.45
GBcorr (T)	.01966	.4819	.4025	.1003	.0231	.4512	.9746	7.00
GBcorr (R)	.01620	.5150	.4247	.0680	.0174	.3031	.6415	4.99

Achim Streit, and Markus Götz, “Differentiable power-flow optimization,” *arXiv preprint arXiv:2603.28203*, 2026.

- [5] Alcir Monticelli, “Electric power system state estimation,” *Proceedings of the IEEE*, vol. 88, no. 2, pp. 262–282, 2000.
- [6] Nan Lin, Stavros Orfanoudakis, Nathan Ordonez Cardenas, Juan S. Giraldo, and Pedro P. Vergara, “PowerFlowNet: Power flow approximation using message passing graph neural networks,” *International Journal of Electrical Power & Energy Systems*, vol. 160, pp. 110112, 2024.
- [7] Olayiwola Arowolo and Jochen L. Cremer, “Towards general-

ization of graph neural networks for AC optimal power flow,” *Energy and AI*, vol. 25, pp. 100842, 2026.

- [8] Luis Piloto, Sofia Liguori, Sephora Madjiheurem, Miha Zgubic, Sean Lovett, Hamish Tomlinson, Sophie Elster, Chris Apps, and Sims Witherspoon, “CANOS: A fast and scalable neural AC-OPF solver robust to $N-1$ perturbations,” *arXiv preprint arXiv:2403.17660*, 2024.
- [9] Tianhao Wen, Wenyue Wang, Jinchang Chen, and Zhaojian Wang, “Power flow surrogate for power systems with high renewable penetration via a physics-informed graph attention network,” *Energies*, vol. 19, no. 13, pp. 2972, 2026.
- [10] Maria Chzhen and Priya L. Donti, “Improving feasibility via fast autoencoder-based projections,” *arXiv preprint arXiv:2604.03489*, 2026.
- [11] Xiang Pan, Minghua Chen, Tianyu Zhao, and Steven H. Low, “DeepOPF: A feasibility-optimized deep neural network approach for AC optimal power flow problems,” *arXiv preprint arXiv:2007.01002*, 2020.
- [12] Hoang T. Nguyen and Priya L. Donti, “FSNet: Feasibility-seeking neural network for constrained optimization with guarantees,” in *Advances in Neural Information Processing Systems*, 2025, vol. 38, pp. 44366–44404.
- [13] Priya L. Donti, David Rolnick, and J. Zico Kolter, “DC3: A

- learning method for optimization with hard constraints,” in *International Conference on Learning Representations*, 2021.
- [14] Weiwei Yang, Andrea Britto Mattos Lima, Thiago V. Spina, Spencer Fowers, Baosen Zhang, and Chris White, “GridSFM: A foundation model for AC optimal power flow,” May 2026, Microsoft Research technical white paper.
- [15] GridFM Team, “gridfm-graphkit: Tools for training, fine-tuning, and interacting with foundation models for the electric power grid,” Python package, version 0.9.0, 2026, Released Aug. 28, 2026; accessed Sep. 11, 2026; source SHA-256: 748fe507c31bb02b937f6eb56298ac3cfb0924304442af439f5f415d0ddbaedf.
- [16] Alban Puech, Matteo Mazzonelli, Tamara R. Govindasamy, Mangaliso Mngomezulu, Héctor Maeso-García, Thomas Tolhurst, Javad Bayazi, Ali Moeini, Naomi Simumba, Celia Cintas, David Nelischer, Romeo Kienzler, Jonas Weiss, Anna Varbella, Florian Dörfler, Gabriela Hug, Martin Mevissen, Juan Bernabé-Moreno, François Mirallès, Hendrik F. Hamann, Etienne Vos, and Thomas Brunschweiler, “GENCO—a unified neural solver embedded in a development framework for steady-state grid analysis,” *arXiv preprint arXiv:2608.09921*, 2026.
- [17] Yijiang Li, Zeeshan Memon, Hongwei Jin, Stefano Fenu, Kunju Song, Sunash B. Sharma, Parfait Gasana, Hongseok Kim, Liang Zhao, and Kibaek Kim, “LUMINA: Foundation models for topology transferable ACOPF,” in *ICLR 2026 Workshop on Foundation Models for Science: Real-World Impact and Science-First Design*, 2026, arXiv:2603.04300.
- [18] Aoxiang Ma, Salah Ghamizi, Aleksandar Bojchevski, Pei Zhang, and Jun Cao, “SafePowerGraph: A safety-aware benchmarking framework for scalable graph neural networks in power flow and optimal power flow,” *Energy and AI*, vol. 25, pp. 100819, 2026.
- [19] Rahul Nellikkath and Spyros Chatzivasileiadis, “Physics-informed neural networks for AC optimal power flow,” *Electric Power Systems Research*, vol. 212, pp. 108412, Nov. 2022.
- [20] Ariel Goodwin, Jonathan Maack, and Devon Sigler, “Power flow geometry and approximation,” *IEEE Transactions on Power Systems*, vol. 41, no. 2, pp. 982–993, 2026.
- [21] Seonho Park, Wenbo Chen, Terrence W. K. Mak, and Pascal Van Hentenryck, “Compact optimization learning for AC optimal power flow,” *IEEE Transactions on Power Systems*, vol. 39, no. 2, pp. 4350–4359, Mar. 2024.
- [22] Ana Rivera Him, Anvita Bhagavathula, Alvaro Carbonero, and Priya L. Donti, “PF Δ : A benchmark dataset for power flow under load, generation, and topology variations,” in *Advances in Neural Information Processing Systems*, 2025, vol. 38, Datasets and Benchmarks Track.
- [23] Changhun Kim, Timon Conrad, Redwanul Karim, Julian Oelhaf, David Riebesel, Tomás Arias-Vergara, Andreas Maier, Johann Jäger, and Siming Bayer, “Physics-informed GNN for medium-high voltage AC power flow with edge-aware attention and line search correction operator,” in *2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026.
- [24] Leon Thurner, Alexander Scheidler, Florian Schäfer, Jan-Hendrik Menke, Julian Dollichon, Friederike Meier, Steffen Meinecke, and Martin Braun, “pandapower—an open-source python tool for convenient modeling, analysis, and optimization of electric power systems,” *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6510–6521, Nov. 2018.
- [25] Ray D. Zimmerman, Carlos E. Murillo-Sánchez, and Robert J. Thomas, “MATPOWER: Steady-state operations, planning, and analysis tools for power systems research and education,” *IEEE Transactions on Power Systems*, vol. 26, no. 1, pp. 12–19, Feb. 2011.
- [26] Samuel Chevalier and Spyros Chatzivasileiadis, “Global performance guarantees for neural network models of AC power flow,” *arXiv preprint arXiv:2211.07125*, 2022.